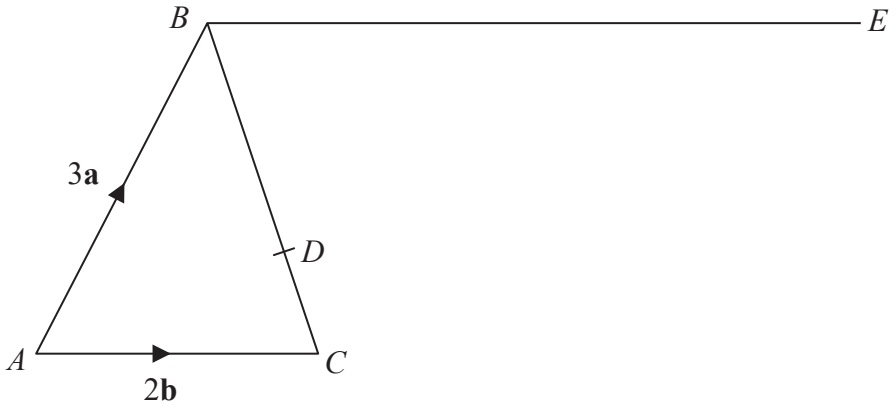
**19**




The diagram shows triangle $ABC$.

$\overrightarrow{AB} = 3\mathbf{a}$

$\overrightarrow{AC} = 2\mathbf{b}$

$\overrightarrow{BE} = 3\overrightarrow{AC}$

$D$ is the point on $BC$ such that $BD : DC = 3 : 1$

Prove that $ADE$ is a straight line.

Vectors: BC = AC − AB = 2b − 3a; BE = 6b.

**1 mark**
BC or BE

AD = AB + ¾BC = 3a + ¾(2b − 3a).

**1 mark**
Vector route to D

= ¾a + 1½b = ¾(a + 2b).

AE = AB + BE = 3a + 6b = 3(a + 2b).

**1 mark**
Correct scaled vectors

Therefore AE = 4AD.

AD and AE are parallel and both start at A.

Hence A,D,E lie on one straight line.

**1 mark**
Parallel; common A

**(Total for Question 19 is 4 marks)**